THÉOPHILE GOUSSELOT

HARDWARE SECURITY / R&D ENGINEER — PHD

Microelectronics · Secure Processor Architectures · RISC-V

I enjoy **exploring hardware from both sides:** understanding how it works and how it can fail, combining processor architecture and RTL design with fault injection and experimental security evaluation.

I love getting into the details, experimenting, scripting, and figuring things out with others — **to develop, attack or protect chips for hours.**



Website ↗
GitHub ↗
theophile.gousselot@gmail.com
CV FR ↗

EXPERIENCE

2022–2025

PhD — Mines Saint-Étienne · PEPR Cybersécurité ARSENE

"Protection of RISC-V processors against fault injection"

- Designed and implemented hardware countermeasures against fault injection and code-extraction attacks (leveraging Texplained's reverse-engineering expertise ↗), from processor architecture and RTL design to FPGA implementation and experimental evaluation.
- Extended Ascon-based instruction encryption by chaining instructions with processor control signals, providing confidentiality and integrity down to the microarchitectural control path.
- Developed a complete verification and characterization infrastructure spanning software compilation/encryption, Verilator simulation, fault-injection modelling, synthesis, post-implementation simulation and FPGA validation.

3rd Hardware Demo Award at IEEE HOST 2023 · San Jose, California

2 peer-reviewed publications · 1 patent

Thesis & full publication list ↗

2021 · 6 months

Hardware Security Engineer — STMicroelectronics

- Designed and implemented a lightweight Control-Flow Integrity (CFI) countermeasure based on CRC signatures for fault-injection detection on a RISC-V processor.
- Developed and evaluated the protection on FPGA by modifying the RTL of a Sodor core, from simulation and synthesis to experimental analysis of instruction-modification detection and theoretical CRC-collision risks.

2020 · 5 months · Montréal

Research Intern — S2MLab

- Developed a generic optimal-control tool in Python/C++ and applied it to optimizing violinist movements.

2019 · 2 months

Engineering Intern — Schneider Electric

- Developed a 3D vision/probing approach for automated defect detection.

TEACHING & LEADERSHIP

Teaching — École des Mines de Saint-Étienne

- Processor Architecture (RISC-V case study) — 22 h/year (2022, 2023, 2024)
- Digital System Design: Ascon implementation — 24 h (2022)

President — MINITEL student network association

Led a 20-student team responsible for the residence network and student IT support under a €5k service contract with the school. (2019)

EXPERTISE

Hardware & Architecture
RISC-V · CPU architecture · RTL design
FPGA · ASIC · SystemVerilog · VHDL

Hardware Security
Fault injection · Physical attacks
Threat modeling · HW countermeasures

Cryptographic Hardware
Hardware implementation of cryptographic primitives · Ascon · AES

Software & Automation
Python · C/C++ · Bash · Linux · Make · Git

Tools
Vivado · Verilator · ModelSim · GTKWave

EDUCATION

2022–2025

PhD in Microelectronics

Mines Saint-Étienne

2018–2021

Graduate Engineer — ISMIN

Mines Saint-Étienne
Microelectronics · Computer Science
Hardware Security

Special Distinction – 3 recipients / 100

2016–2018

Preparatory Class

Mathematics & Physics

LANGUAGES

French Native
English C1

BEYOND ENGINEERING

Theatre acting · Climbing · Team sports
Getting Things Done fan · Vim geek

Europe by bike
10,000 km · 363 days · 2025-2026
Fully self-supported